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Gursirat Singh

Software Engineer | Full Stack Developer

Portfolio: gursirat-portfolio.vercel.app
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Computer Science undergraduate at Lovely Professional University with a strong foundation in Data Structures & Algorithms (LeetCode Rating: 1991) and experience building scalable full-stack web applications using Next.js, React.js, TypeScript, Node.js, Express.js, MongoDB, and RESTful APIs. Passionate about building scalable full-stack web applications, developing backend services with Node.js, creating responsive React.js interfaces, and delivering production-ready software using modern cloud technologies.

SKILLS

Languages: Java, JavaScript, TypeScript, Python, C++

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS, Responsive Web Design

Backend: Node.js, Express.js, RESTful APIs, JWT Authentication

Databases: MongoDB, PostgreSQL, MySQL, Redis

Cloud & DevOps: AWS (EC2, S3, IAM), Vercel, Git, GitHub

AI & LLM: LangChain.js, LangGraph.js, Gemini API, Prompt Engineering, Structured Output (Zod)

PROJECTS

VestPulse | Next.js, TypeScript, LangGraph.js, Gemini API, Upstash Redis

Jul 2026

[GitHub](#)

[Live Demo](#)

- Built a full-stack AI investment research platform using Next.js, TypeScript, LangGraph.js, Gemini API, and RESTful APIs with a 7-node fan-out/fan-in workflow that parallelized financial, news, competitor, and risk analysis, reducing pipeline latency by 60%.
- Optimized backend service orchestration by consolidating seven sequential Gemini API calls into a single structured-output workflow using Zod validation, reducing external API usage by 85% while improving application performance and reliability.
- Secured REST API integrations through schema validation, Redis-backed rate limiting (5 requests/min/IP), and input verification, improving application security and reliability.
- Deployed the application on Vercel with multi-provider financial data failover, client-side PDF export, responsive UI, and Lighthouse scores of 99 Performance, 100 Accessibility, 100 Best Practices, and 100 SEO.

InnoPulse | Next.js, TypeScript, MongoDB, Node.js

Oct 2025 - Apr 2026

[GitHub](#)

[Live Demo](#)

- Developed a responsive full-stack startup discovery platform using Next.js, Node.js, MongoDB, and RESTful APIs with dynamic multi-filter search, role-based access control, and an admin approval workflow for 50+ startups.
- Architected a Puppeteer-based PDF generation microservice deployed on Render, eliminating Vercel serverless timeout limitations and reducing PDF generation time to under five seconds.
- Automated startup data synchronization through scheduled Vercel Cron Jobs, eliminating manual updates across 50+ startup records and improving data consistency.
- Optimized MongoDB queries and Next.js API routes, reducing multi-filter search latency from nearly one second to under 300 milliseconds and improving overall application responsiveness.

EDUCATION

Bachelor of Technology in Computer Science and Engineering

Lovely Professional University, Phagwara, Punjab
CGPA: 7.84

Aug 2023 – Present

Intermediate (PCM)

Swami Sant Dass Public School, Jalandhar, Punjab
Percentage: 83%

Mar 2022 – May 2023

Matriculation

Swami Sant Dass Public School, Jalandhar, Punjab
Percentage: 87%

Mar 2020 – May 2021

CERTIFICATES

Cyber Threat Management

Cisco

Apr 2026

Cloud Computing

NPTel

Oct 2025